A380X OIS and ATC datalink system model

The IMA / datalink model the OIS applications build on: ATC MSG lights and RING inhibited in takeoff and landing with the uplinks rejected then, the ARU start of the FLT OPS network (STANDALONE), the OIS MENU and OIS XFR keys (X-OIS), COMPANY COM weather request and free text, EXIT SESSION, printing a COMPANY COM message, the VHF 3 and SCI failures that stop the datalink and the AOC, and the AOC hosted by the NSS AVNCS ANSUs with NSS DATA TO AVNCS OFF stopping its downlinks

Aircraft	A380X
Where to find it	The OITs (FLT OPS and NSS AVNCS sides), the laptop keyboards (OIS MENU, OIS XFR), the ATC MSG pbs and the SD ATC mailbox; the overhead NSS panel (NSS DATA TO AVNCS); flyPad > Failures (ATA 23, 46)
Branch	`core/a380x-ois-datalink`, based on `master` (FlyByWire master 76db0aef1) with `core/printer`, `core/a380x-mailbox-du-reconf` and `core/a380x-elec` merged in: the AOC print goes through the FMS printer of core/printer, the ATC MSG lights of core/a380x-mailbox-du-reconf, the ARU start reads the ANSUs of core/a380x-elec. `core/a380x-ois` is stacked on it. It holds the OIS and ATC mailbox parts of the IMA/datalink model (83f832d77), the OIS / AOC part of 30af1a232 with its systems host side (the AOC on the ANSUs), and the datalink part of 986919394 (VHF 3 and SCI failures), from the `core/fcom-gap-fixes` archive and develop. Left out: the NSS smoke failures and SMOKE light of 83f832d77 (on `core/a380x-elec-fuel-alerts` and `core/failures-a32nx`, with the smoke detection), the GPWS / TCAS priorities, WXR knobs and relight EGT of 30af1a232 (on `core/a380x-elec-fuel-alerts`, `core/a380x-wxr-controls`, `core/failures-a380x` and `core/failures-a32nx`), its ATC RESET part, and the FWS avionics network / ATA 46 COM alerts and ADS failure 46010 of 986919394 (on `core/a380x-elec-fuel-alerts`).
Status	Not sim-tested on this branch. On develop the OIS MENU and OIS XFR keys (X-OIS) were seen working in the sim with the ACD keys (2026-10-17) and the OIS applications built on this model passed their sim tests (2026-10-18). The ATC MSG inhibition and uplink rejection: vitest specs only. VHF 3 / SCI failures and the AOC on the ANSUs (placed 2026-10-18): not sim-tested on the core branch; unit tests only (on develop they were in the OIS install of 2026-10-18 but not tested on their own). The build configuration commit c49018f8a (OIT, SDv2 and extras host rebuilt when their outside sources change) has no user-visible effect. AOC off receives nothing (92a4d3b28, 2026-10-21): sim-tested on develop in the OIS batch 2 install (T7, T12), not on the core branch.
Pictures	No pictures: the X-OIS message, STANDALONE and the forms are small additions to stock OIT pages.

1. ATC MSG lights and RING inhibited in takeoff and landing
2. ARU start: STANDALONE and SYNCHRONIZATION NOT AVAIL
3. OIS MENU and OIS XFR keys (X-OIS)
4. COMPANY COM weather request and free text
5. EXIT SESSION
6. Printing a COMPANY COM message
7. VHF 3 failure stops the datalink; SCI 1 and SCI 2 failures
8. AOC hosted by the NSS AVNCS ANSUs; NSS DATA TO AVNCS OFF stops the AOC downlinks

1. ATC MSG lights and RING inhibited in takeoff and landing

> **Aircraft:** A380X | **Where:** Glareshield ATC MSG pbs, ATC mailbox on the SD | **Status:** Not sim-tested on this branch; vitest specs pass (check-only OIT, MFD, SystemsHost, SDv2).

What it is for

During takeoff and landing the ATC MSG pb lights and the RING stay off. An uplink received then, or a sixth uplink that needs an answer while five already wait for one, is rejected: the ATC center gets an automatic ERROR downlink. A message that already waits flashes again after the phase.

What you should see

Situation	What you should see
An uplink waits, FWS phase 3 to 7 (takeoff power to 1500 ft) or 9 to 10 (800 ft to 80 kt)	ATC MSG lights dark, no RING.
A new uplink arrives in those phases	It is not shown; the center receives ERROR.
After the phase, a message still waits	The ATC MSG lights flash again.

Limitations and why

Limitation	Why	What to do
Taxi is not inhibited.	Design choice: the manual names taxiing in one sentence only; inhibiting the taxi would reject the taxi and departure clearances. Same phases as the COMPANY MSG memo inhibition.	None.
A32NX unchanged.	The rules are behind the A380X mailbox flag of the shared ATC datalink.	None.

How to use

1. Nothing to do: the inhibition follows the FWS flight phase.

How to test

#	Do this	Expected result
1	Hoppie connected, logged on: have the center send a message during the takeoff roll.	No ATC MSG light, no RING; the center receives an ERROR message.
2	Have a message sent before takeoff power and leave it unread through the takeoff.	The lights stop at takeoff power and flash again above 1500 ft.

2. ARU start: STANDALONE and SYNCHRONIZATION NOT AVAIL

> **Aircraft:** A380X | **Where:** OIT FLT OPS side: Login page and FLT OPS STS page | **Status:** Not sim-tested on this branch; vitest specs pass (check-only OIT, MFD, SystemsHost, SDv2).

What it is for

The laptops connect to the FLT OPS network through the ARU, which operates 2 minutes after the FLT OPS ANSU is powered. A laptop started before that, or without the network, works as a stand-alone: STANDALONE on its Login page and SYNCHRONIZATION NOT AVAIL on the FLT OPS STS page.

Limitations and why

Limitation	Why	What to do
The ARU follows the FLT OPS ANSU.	Design choice: the ARU supply and failures are not modelled.	None.
A laptop started before the ARU stays a stand-alone.	As the manual: restart it once the ARU operates.	Log off and start the laptop again.

How to use

1. After a power-up, wait 2 minutes before starting the laptops.

How to test

#	Do this	Expected result
1	Cold and dark, power up, start the FLT OPS laptop at once.	Login page: STANDALONE.
2	Wait 2 minutes, restart the laptop.	No STANDALONE; FLT OPS STS without SYNCHRONIZATION NOT AVAIL.

3. OIS MENU and OIS XFR keys (X-OIS)

> **Aircraft:** A380X | **Where:** Laptop keyboards (and the ACD keys with core/a380x-cockpit): OIS MENU, OIS XFR | **Status:** Seen working in the sim on develop with the ACD keys (2026-10-17). Not sim-tested on this branch; vitest specs pass (check-only OIT, MFD, SystemsHost, SDv2).

What it is for

OIS MENU shows the FLT OPS menu. OIS XFR starts X-OIS: the OIT shows the FLT OPS page of the other OIT with "X-OIS Activated", and follows it; OIS XFR again (or OIS MENU) ends it.

Limitations and why

Limitation	Why	What to do
X-OIS shows the other OIT's page, with this OIT's own laptop data.	Design choice: the two OITs share only the page code (L:A380X_OIT_CAPT/FO_FLT_OPS_PAGE); the synced bus does not reach the other OIT of the same VCockpit.	None.
The ACD keys need core/a380x-cockpit.	The ACD legends and click areas are there.	Use the laptop keyboard keys.

How to use

1. Press OIS XFR to see the other side's FLT OPS page; press it again to come back.

How to test

#	Do this	Expected result
1	F/O OIT on T.O PERF, CAPT OIT on the menu: CAPT presses OIS XFR.	CAPT OIT shows T.O PERF with "X-OIS Activated".
2	CAPT presses OIS XFR again.	CAPT OIT back on its menu.

4. COMPANY COM weather request and free text

> **Aircraft:** A380X | **Where:** NSS AVNCS > COMPANY COM | **Status:** Not sim-tested on this branch; vitest specs pass (check-only OIT, MFD, SystemsHost, SDv2).

What it is for

COMPANY COM gets a METAR / TAF request page and a free text page, sent through the AOC (Hoppie network); the answers come into the Inbox.

Limitations and why

Limitation	Why	What to do
No NOTAM request.	The datalink has no NOTAM source.	None.

How to use

1. COMPANY COM > Requests > Weather: enter the airports, tick METAR / TAF, SEND.

How to test

#	Do this	Expected result
1	Hoppie connected: request the METAR of the departure airport.	The answer arrives in the Inbox.

5. EXIT SESSION

> **Aircraft:** A380X | **Where:** OIT FLT OPS MENU page | **Status:** Not sim-tested on this branch; vitest specs pass (check-only OIT, MFD, SystemsHost, SDv2).

What it is for

EXIT SESSION returns to the Login page and clears the FLT OPS STS page for the next flight. It is not available while the FLT FOLDER is open.

Limitations and why

Limitation	Why	What to do
None known.	-	-

How to use

1. Close the FLT FOLDER, then FLT OPS MENU > EXIT SESSION.

How to test

#	Do this	Expected result
1	Enter a flight number on FLT OPS STS, then EXIT SESSION.	Login page; after the login FLT OPS STS is empty.
2	Open the FLT FOLDER, back to the menu.	EXIT SESSION greyed.

6. Printing a COMPANY COM message

> **Aircraft:** A380X | **Where:** NSS AVNCS > COMPANY COM > Inbox, pedestal printer | **Status:** Not sim-tested on this branch; vitest specs pass (check-only OIT, MFD, SystemsHost, SDv2).

What it is for

PRINT under a displayed Inbox message prints it (subject, sender, reception time, text) on the pedestal printer, through the FMS printer of the MFD. The AOC also gets a downlink inhibition (no messages sent, uplinks still received), which the NSS DATA TO AVNCS pb-sw drives (see "AOC hosted by the NSS AVNCS ANSUs" below).

Limitations and why

Limitation	Why	What to do
No print while the printer is off or unpowered.	The FMS printer answers PRINTER NOT AVAIL.	Switch the printer on.

How to use

1. Inbox > select a message > PRINT.

How to test

#	Do this	Expected result
1	Receive a company message, open it in the Inbox, PRINT.	The pedestal printer prints it.
2	No message selected.	PRINT greyed.

7. VHF 3 failure stops the datalink; SCI 1 and SCI 2 failures

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 23 Communication and ATA 46 Information Systems | **Status:** Not sim-tested on the core branch. Develop (986919394): unit tests only (A380xVhfProvider.spec.ts, AocAvailability.spec.ts), no sim test recorded.

What it is for

- VHF 3 is the only data medium of the A380X datalink router (no HF or SATCOM datalink in the sim). A failed VHF 3 (flyPad failure **VHF 3**, id 23007) is now treated as not in DATA mode: CPDLC and AOC stop, as COM VHF 3 DATALINK FAULT tells the crew on the aircraft. Before, the datalink kept working with VHF 3 failed.
- Two new flyPad failures: **SCI 1 (NSS / avionics secure communication interface)** and **SCI 2 (NSS / avionics secure communication interface)** (46008, 46009). With both lost the AOC is inoperative: it can no longer send or receive messages (A380 FCOM COM NSS/AVNCS DATA XFR FAULT). One SCI alone changes nothing.

Limitations and why

Limitation	Why	What to do
No ECAM alert: no COM VHF 3 DATALINK FAULT, no COM NSS/AVNCS DATA XFR FAULT.	The FWS avionics network and ATA 46 COM alerts (and the ADS function failure 46010) extend develop alert code that is on no core branch; they stay on develop.	Know which failure you set.

How to use

1. flyPad > Failures > ATA 23 Communication: **VHF 3** to lose the datalink.
2. flyPad > Failures > ATA 46 Information Systems: **SCI 1** and **SCI 2** together to lose the AOC.

How to test

#	Do this	Expected result
1	A380X powered, Hoppie connected, VHF 3 in DATA mode on the RMP. OIT COMPANY COM: request the METAR of the departure airport.	The answer arrives in the Inbox.
2	flyPad > Failures > ATA 23: **VHF 3**. Request the METAR again.	No answer: the request fails. Deactivate the failure: requests work again.
3	flyPad > Failures > ATA 46: **SCI 1** only. Request the METAR.	The answer arrives.

4	Add **SCI 2**. Request the METAR.	No answer: the AOC is off. Deactivate both: the AOC works again.

8. AOC hosted by the NSS AVNCS ANSUs; NSS DATA TO AVNCS OFF stops the AOC downlinks

> **Aircraft:** A380X | **Where:** NSS DATA TO AVNCS pb-sw and NSS MASTER sw (overhead NSS panel); flyPad > Failures > ATA 46 | **Status:** Not sim-tested on the core branch. Develop (30af1a232): unit tests only (AocAvailability.spec.ts), no sim test recorded.

What it is for

- The AOC ran while AC BUS 1 was powered. It now runs while one NSS AVNCS ANSU is healthy: "In the case of an ANSU 1 failure, ANSU 2 provides the AOC application" (A380 FCOM DSC-46-40-10-20). An ANSU is healthy when powered, not failed (flyPad **NSS AVNCS ANSU 1** / **NSS AVNCS ANSU 2**), not reset (RESET panel NSS AVNCS) and with the NSS MASTER sw on; the OIT writes it (L:A32NX_NSS_ANSU_1(2)_IS_HEALTHY). With both SCIs lost the AOC stays off.
- NSS DATA TO AVNCS pb-sw OFF (pressed in, OFF light) stops the AOC downlinks (messages and requests); the uplinks still come in (FCOM DSC-46-20-20-80). The systems host reads the pb-sw itself: 1 = normal, 0 = OFF.
- While the AOC is off it receives no company message (a free text the ground sends then never reaches the aircraft), and no COMPANY MSG stays after it powers down (92a4d3b28, the AOC part of develop 1e9ee7910; fbw-common, the A32NX AOC behaves the same).

Limitations and why

Limitation	Why	What to do
The AOC follows the ANSU health written by the OIT.	The ANSUs are modelled in the OIT instrument; the systems host reads their L:vars twice a second.	-
No COM NSS/AVNCS DATA XFR FAULT or AVIONICS NETWORK alert.	Those FWS alerts are on `core/a380x-elec-fuel-alerts`; merge it to get them.	-

How to use

1. Keep the NSS MASTER sw on and the NSS DATA TO AVNCS pb-sw in its normal position for the company datalink.
2. Press NSS DATA TO AVNCS (OFF) to stop sending to the ground while still receiving.

How to test

#	Do this	Expected result
1	A380X powered, Hoppie connected, VHF 3 DATA. OIT COMPANY COM: request the METAR of the departure airport.	The answer arrives in the Inbox.
2	Press NSS DATA TO AVNCS (OFF light). Request the METAR again.	Nothing is sent: no answer. Release the pb-sw: requests work again.
3	flyPad > Failures > ATA 46: **NSS AVNCS ANSU 1**. Request the METAR.	The answer arrives (ANSU 2 hosts the AOC).
4	Add **NSS AVNCS ANSU 2**. Request the METAR.	No answer: the AOC is off. Deactivate both.
5	With an unread company message, fail both ANSUs.	The COMPANY MSG memo goes away and does not come back when the AOC runs again.